THE NOVA RESEARCH CURRICULUM

General Science

Grade 6 · Term 1

Student edition

Outcomes: SC-6.1 cells and the microscope, SC-6.2 reproduction in plants, SC-6.3 digestion and a balanced diet.

Lesson notes

SC-6.1 Cells (Weeks 1–4)

Cells are the basic unit of life. Animal cell: nucleus, cell membrane, cytoplasm, mitochondria. Plant cell also has a cell wall, chloroplasts and a large vacuole. Practical: onion-skin cells under a microscope or a teacher's microscope video; draw and label. [picture: plant and animal cells] [simulation: cell explorer]

SC-6.2 Reproduction in plants (Weeks 5–8)

Sexual reproduction (flowers, pollination, fertilisation, seeds) and asexual reproduction (runners, bulbs, cuttings). Practical: grow a money-plant cutting and a seed side by side.

SC-6.3 Digestion and diet (Weeks 9–12)

Mouth (saliva and amylase) → oesophagus → stomach → small intestine (absorption) → large intestine (water). Nutrients: carbohydrates, proteins, fats, vitamins, minerals, fibre, water; deficiency disorders (scurvy, anaemia). Practical: test starch with iodine. [picture: digestive system]

Workbook pages

Page	Practical or task
1	Microscope drawing of onion cells with labels.
2	Compare plant and animal cells in a table.
3	Cutting vs seed growth record (4 weeks).
4	Label the digestive system; trace a meal's journey.
5	Starch test practical report.

Question bank

#	Outcome	Question	Your answer
1	SC-6.1	Which part controls the cell?	cell wall / nucleus / vacuole / cytoplasm
2	SC-6.1	Only plant cells have chloroplasts. True or false.	True / False
3	SC-6.1	Name two parts found in plant cells but not animal cells.	_____

#	Outcome	Question	Your answer
4	SC-6.2	Growing a plant from a cutting is ... reproduction.	sexual / asexual
5	SC-6.2	Seeds form after fertilisation. True or false.	True / False
6	SC-6.3	Where is most food absorbed?	stomach / small intestine / large intestine / mouth
7	SC-6.3	Saliva contains an enzyme that starts digesting ...	protein / fat / starch / water
8	SC-6.3	Lack of vitamin C causes ...	anaemia / scurvy / rickets / goitre
9	SC-6.3	Iodine turns blue-black when starch is present. True or false.	True / False
10	SC-6.3	Plan a balanced lunch and name one nutrient from each food.	_____